Student's assessment number.....

THE UNITED REPUBLIC OF TANZANIA

PRIME MINISTER'S OFFICE REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM TWO PRE-NATIONAL EXAMINATION

COMPUTER SCIENCE

Time: **20:30Hours** **Year:2026**

Instructions

1. This paper consists of Section A, B and C with a total of ten (10) questions.
2. Answer **all** questions in the space provided
3. Section A and C carry **fifteen (15)** marks each and section B carries seventy (70) marks
4. All answer must be written in **black** or **blue** ink pen.
5. Communication devices and any other unauthorized materials are not allowed in the examination room.
6. Write your **Assessment number** at the top carrier of every page.

FOR ASSESSORS ONLY		
QUESTION NUMBER	**SCORE**	**ASSESSOR'S INITIAL'S**
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		
CHECKERS INITIALS		

SECTION A:(15 Marks)

Answer all questions from this section

1. For each of the items (i)-(x) choose the correct answer from the given alternatives and write its letter in the box provided.
   i. Select the true statement about array in C
      A. Arrays can store different types of data in C
      B. Arrays are used to store multiple values value of the same data type
      C. Arrays in C dynamic sizes by default
      D. Arrays in C are always initialized to zero
   ii. Laptop and desktops are examples of which classification of computers?
      A. Supercomputers
      B. Minicomputers
      C. Microcomputers
      D. Mainframes
   iii. What is an IP address, and why is it important for internet communication?
      A. An IP address is a password used to access websites
      B. An IP address is a unique identifier for devices that allows communication between them over the internet
      C. An IP address is a type of encryption used for security data
      D. An IP address is only needed for mobile data connections
   iv. Which of the following is a preventive maintenance practices for computer hardware?
      A. Deleting system files
      B. Cleaning dusts from internal components
      C. Ignore software updates notifications
      D. Uninstalling the anti-virus
   v. A user receives an email claiming they have win a lottery and are asked to click a link to claim their prizes. This is a classic example of which type of cyber threat?
      A. Malware
      B. Phishing
      C. A Distributed Denial Service (**DDoS**) attack
      D. A Trojan Horse
   vi. Which computer generation introduces the concept of personal computer (PCs) and graphical user interface?
      A. Second Generation
      B. Third Generation
      C. Fourth Generation
      D. Fifth Generation
   vii. In C programming which escape sequence is used to move the cursor in a new line?
      A. \t
      B. \a

C. \n
D. \b

viii. Which hardware component is often referred to as the "brain of the computer"?
A. RAM
B. Hard Drive
C. Motherboard
D. CPU

ix. What is the correct order of the first three steps in the problem -solving cycle?
A. Design, Define, Analyse
B. Define, Analyse, Design
C. Code, Test, Design
D. Analyse, Design, Code

x. Which task would be best solved using a loop in a program?
A. Typing your name once
B. Printing the same message 100 times
C. Saving a single
D. Opening one folder

Answers.

LIST A	**(i)**	**(ii)**	**(iii)**	**(iv)**	**(v)**	**(vi)**	**(vii)**	**(viii)**	**(ix)**	**(x)**
LIST B										

2. Match the following Programming concepts in **LIST A** with their correct description in **LIST B**

LIST A	**LIST B**
i. Speed ii. Diligence iii. Accuracy iv. Storage Capability v. versatility	A. computer perform different task B. perform calculations and give correct results C. A storage location for changing data D. How quickly a computer completes tasks E. Large number of data is stored and retrieved F. Computer does not get tired of fatigued

Answers.

LIST A	**(i)**	**(ii)**	**(iii)**	**(iv)**	**(v)**
LIST B					

SECTION B:(70% Marks)

Answer all questions from this section

3. a) Explain how the internet can help students learn better both in school and at home

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

b) Briefly explain three advantages and two disadvantages of E-learning

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

4. a) In your own words, explain five importance of computer science

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

b) (i)Give three why a mobile phone is regarded as computer

.....
.....
.....
.....
.....

(ii)Explain two roles of the motherboard in computer system and how it connects various hardware components.

.....
.....
.....
.....
.....

5. a) (i) Define the term:

Cybersecurity.....
.....
.....
.....
.....

(ii)How do common cyberattacks impact individuals and business in Tanzania today?

.....
.....
.....
.....
.....

b) (i) What is a **Cloud Security**? Why is it a crucial aspect of safeguard your data?

.....
.....
.....
.....
.....
.....
.....
.....
.....
.....

(ii) Briefly explain four best ways, how can protect your password.

.....

.....

.....

.....

.....

.....

.....

.....

6. a) (i) Briefly explain two main types of computer software

.....

.....

.....

.....

(ii) Give three importance of computer handling and care

.....

.....

.....

.....

.....

.....

b) (i) Why is it necessary to perform computer maintenance in clean workspace?

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) If a printer is not responding to print requests, explain the steps you would take to resolve the problem.

.....

.....

.....

.....

.....

.....

.....

.....

7. a) (i) What is **Computer Programming**?

.....

.....

.....

(ii) Why is computer programming important in today's world?

.....

.....

.....

.....

.....

.....

.....

.....

.....

b) (i) Define the term **Python**

.....

.....

.....

(ii) Differentiate between **Machine Language** and **Assembly Language** (Give three differences

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

8. a) (i) What is an **Algorithm**?

.....

.....

.....

(ii) Briefly explain four importance of an algorithm

.....

.....

.....

.....

.....

.....

b) (i) Define the term **Flowchart**

.....

.....

.....

(ii) Write an algorithm for calculating the sum of any two numbers an entered by the user and then draw a flowchart for it.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

9. a) (i) The following C program has a compilation error. Identify the error and correct it

```
#include   <stdio.h>
int main () {
        printf ("Hollo, world\n")
 return 0;
   }
```

.....
.....
.....
.....
.....
.....
.....
.....
.....

(ii) Briefly explain four common Data type in C program

.....
.....
.....
.....
.....
.....
.....
.....

b) What is the output of the following code?

```
#include   <stdio.h>
int main () {
        int x=10;
        float y=3.14;
        printf ("%d\n",   x);
        printf ("%.2f\n", y);
return 0;
   }
```

.....
.....
.....
.....

SECTION C:(15% Marks)

10. a) What is the correct data type for each of the following variables?

(i) Age of a person: .....

(ii) Weight of a person: .....

(iii) Initial of a name.....

b) Write a C program that declares three variables, one **int**, one **float**, and one **char**. Initialize them with appropriate values and print them out.